Republic of the Philippines
Department of Environment and Natural Resources
MINES AND GEOSCIENCES BUREAU
North Avenue, Diliman, Quezon City, Philippines
Tel. No. (+63 2) 8928-8642/ 8928-8937 Fax. No. (+63 2) 8920-1635 E-mail: central@mgb.gov.ph

FLOOD THREAT ADVISORY

To: Brgy. Chairperson
Brgy. Santa Elena
City of Marikina
Metropolitan Manila

Date: October 9, 2024

Dear Sir/Madam:

Please be advised that the Geohazards Mapping and Assessment Team (GMAT) of the Mines and Geosciences Bureau (MGB) has conducted **flood assessment** for the updating of 1:10,000 Scale Geohazard Map in your barangay. The following are the results and recommendations following the assessment:

Purok / Sitio / Area of Concern	GPS Coordinates (Luzon 1911)	Flood Susceptibility Rating	Geohazards Assessment
Barangay Hall and vicinity	14° 38' 04.3" N, 121° 05' 45.1" E	Very High	The barangay hall was inundated with greater than 2m flood height during Ondoy, while ~1m was reported during Ulysses. Floodwaters during said tropical cyclones subsides overnight.
Lambak / Riverside	14° 37' 56.2" N, 121° 05' 34.8" E	Very High	Area is adjacent to the Marikina River. During Ondoy, the first floor was completely submerged midway along the sloping road, while the 2nd floor of buildings closest to the river were also inundated. In recent years floodwaters rarely reached the jogging lane along the river and the river water level recedes fast. No observed dike or flood mitigation along river, but there is ongoing piling in some sections. Moreover, evidence of riverbank erosion manifested as tilted riprap along the Jesus Dela Peña side of the Marikina River.
Mountainview	14° 38' 00.8" N, 121° 06' 11.2" E	Very High	A creek traverses this portion of Mountainview, which causes inundation of nearby residential areas of greater than 2m-high floodwaters, as reported during Ondoy.

Remarks / Recommendations:

- ✓ It is highly recommended to implement appropriate easements in construction of settlements or other infrastructure along rivers/creeks as specified in the Presidential Decree No. 1067 known as Water Code of the Philippines.
- ✓ Improvement and proper maintenance of existing drainage system in the barangay are recommended to prevent inundation of residential areas, critical facilities, and other assets. Avoid improper waste disposal, particularly along drainage canals as it may cause clogging and obstruction of water flow. Ensure regular clearing of natural and artificial drainage pathways, or prior to an impending tropical cyclone or other rain-bearing weather systems.
- ✓ Construction of appropriate and well-studied engineering measures (e.g., dike, flood wall, etc.) should be implemented to minimize the rate of erosion along rivers/creeks. Ensure that the structure follows engineering standards and has structural integrity. Also, damages in the structural mitigation measures should be reported to concerned agency/office responsible for the installation.

The LGU is also encouraged to maintain such measures or reinforce identified damages, with the strong support of the barangay officials and active engagement of the communities and the private sector.

- ✓ Install a warning signage alerting motorists and residents of the danger of crossing bridges susceptible to flash flood during heavy rainfall events.
- ✓ The CDRRMO, together with the BDRRMO, should conduct Information, Education, and Communication (IEC) Campaign on flood hazards to flood-prone communities, particularly the households residing proximal to the river and irrigation canals.
- ✓ Strengthen and improve flood warning system with corresponding warning levels for pre-emptive evacuation purposes. The barangay should have constant communication with nearby and/or upland/upstream barangays during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas.
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of tropical cyclones and other prevailing weather disturbances. This may include City Disaster Risk Reduction Management Office (CDRRMO), Office of the Civil Defense (OCD) NCR, and other related agencies.

Prepared by:

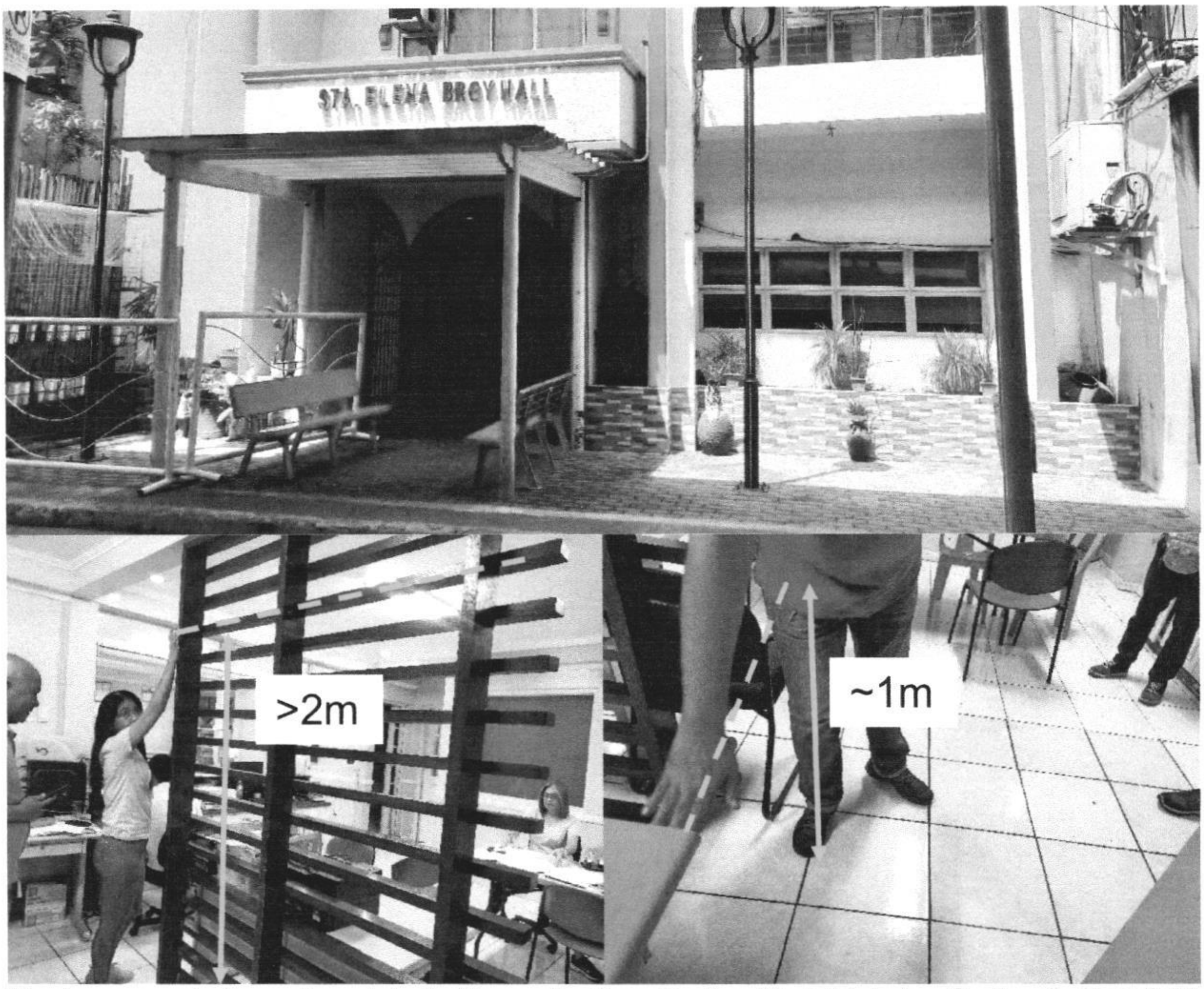
JULIUS VINCENT UMALI
Senior Geologist

MA. LOURDES STEPHANIE V. LEIDO
Senior Geologist

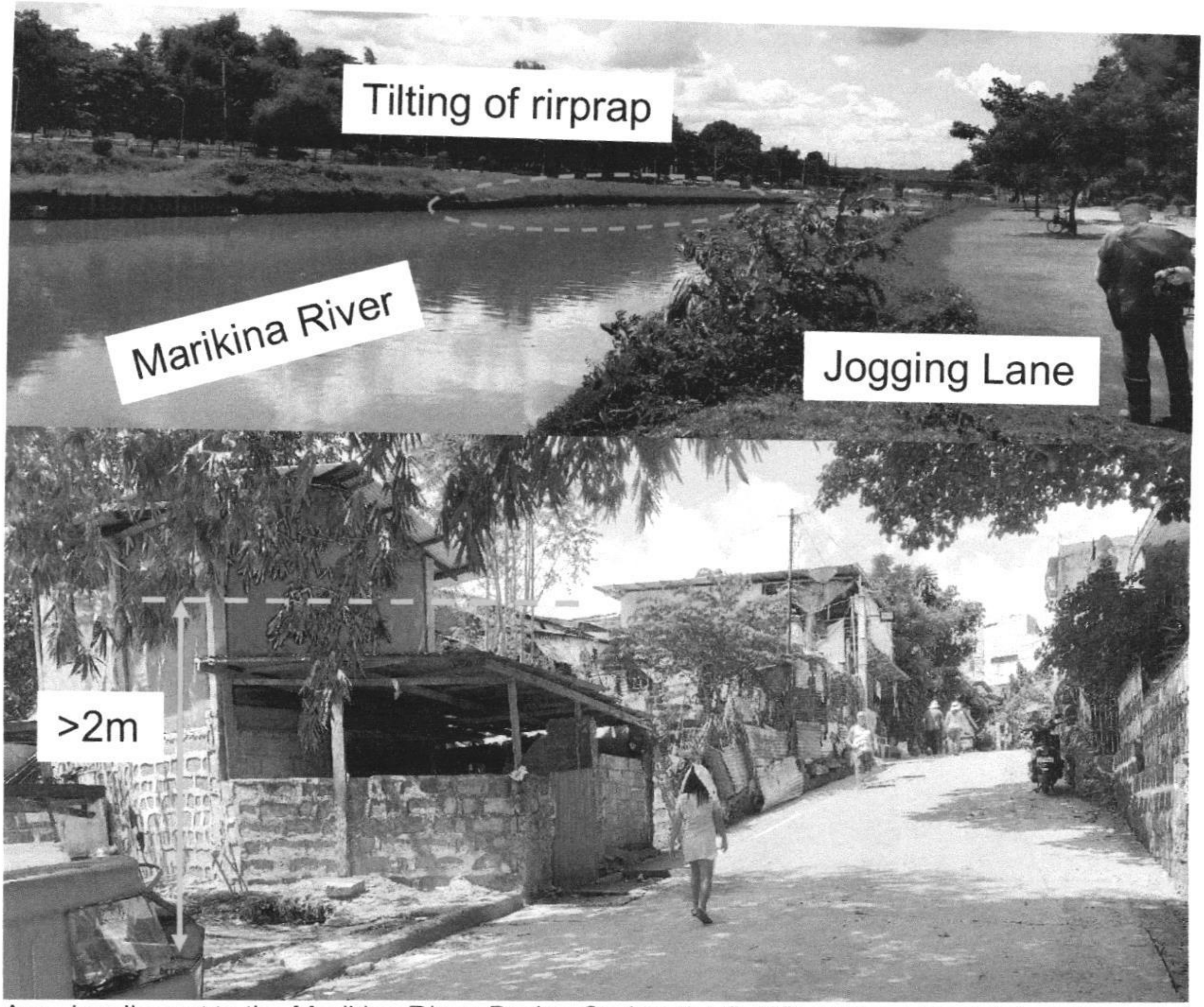
JETHRO B. CAPINO
Geologist II

Received by:
Designation:
Office:

BARANGAY SANTA ELENA



The barangay hall was inundated with greater than 2m flood height during Ondoy *(bottom left)*, while ~1m was reported during Ulysses *(bottom right)*. Floodwaters during said tropical cyclones subsides overnight (GPS Coordinates: 14° 38' 04.3" N, 121° 05' 45.1" E).



Area is adjacent to the Marikina River. During Ondoy, the first floor was completely submerged midway along the sloping road, while the 2nd floor of buildings closest to the river were also inundated. In recent years floodwaters rarely reached the jogging lane along the river and the river water level recedes fast. No observed dike or flood mitigation along river, but there is ongoing piling in some sections. Moreover, evidence of riverbank erosion manifested as tilted riprap along the Jesus Dela Peña side of the Marikina River (GPS Coordinates: 14° 37' 56.2" N, 121° 05' 34.8" E).

A creek traverses this portion of Mountainview, which causes inundation of nearby residential areas of greater than 2m-high floodwaters, as reported during Ondoy (GPS Coordinates: 14° 38' 00.8" N, 121° 06' 11.2" E).

Recommendations:

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